

分数阶傅里叶变换 (FrFT) 原理与应用

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2025.12.25

提 纲

从FT到FrFT

FrFT的基本原理

FrFT的基本性质

FrFT的应用设想

从FT到FrFT

傅里叶变换(FT) —— 积分算符

$$(\mathcal{F}x)(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$

积分算符 —— 算符本征值问题

$$\mathcal{F}\{\psi_n(t)\} = \lambda_n \psi_n(t)$$

Hermite-Gaussian多项式

$$\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi}2^n n!}} e^{-x^2/2} H_n(x) = \frac{(-1)^n}{\sqrt{\sqrt{\pi}2^n n!}} e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2}$$

$$\lambda_n = e^{-in\pi/2}$$

$$\mathcal{F}\{\psi_n(t)\} = \lambda_n \psi_n(t)$$

↓ 作用N次 (整数)

$$\mathcal{F}^N\{\psi_n(t)\} = \lambda_n^N \psi_n(t)$$

↓ 作用p次 (实数) ?

$$\mathcal{F}_p \psi_n(t) = \lambda_n^p \psi_n(t) = e^{-ipn\pi/2} \psi_n(t)$$

分数阶傅里叶变换 本征值形式

基本原理

作用于本征函数

$$\mathcal{F}_p \psi_n(t) = \lambda_n^p \psi_n(t)$$

作用于一般函数

$$\begin{aligned} \mathcal{F}_p[f](x) &= \sum_{n=0}^{\infty} c_n e^{-ipn\pi/2} \psi_n(x) \\ &= \sum_{n=0}^{\infty} \left(\int_{-\infty}^{\infty} f(t) \psi_n(t) dt \right) e^{-ipn\pi/2} \psi_n(x) \\ &= \int_{-\infty}^{\infty} f(t) \left[\sum_{n=0}^{\infty} e^{-ipn\pi/2} \psi_n(x) \psi_n(t) \right] dt \end{aligned}$$

本征函数展开

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} c_n \psi_n(x) \\ c_n &= \int_{-\infty}^{\infty} f(t) \psi_n(t) dt \end{aligned}$$

$$\alpha = p\pi/2 \quad K_\alpha(x, t) = \sqrt{\frac{1 - i \cot \alpha}{2\pi}} \exp \left[i \frac{x^2 + t^2}{2} \cot \alpha - i \frac{xt}{\sin \alpha} \right]$$

分数阶傅里叶变换 积分算符形式

$$\mathcal{F}_p[f](x) = \int_{-\infty}^{\infty} K_p(x, t) f(t) dt$$

基本原理

本征值 $\mathcal{F}_p \psi_n(t) = \lambda_n^p \psi_n(t)$

$\downarrow p = 1$

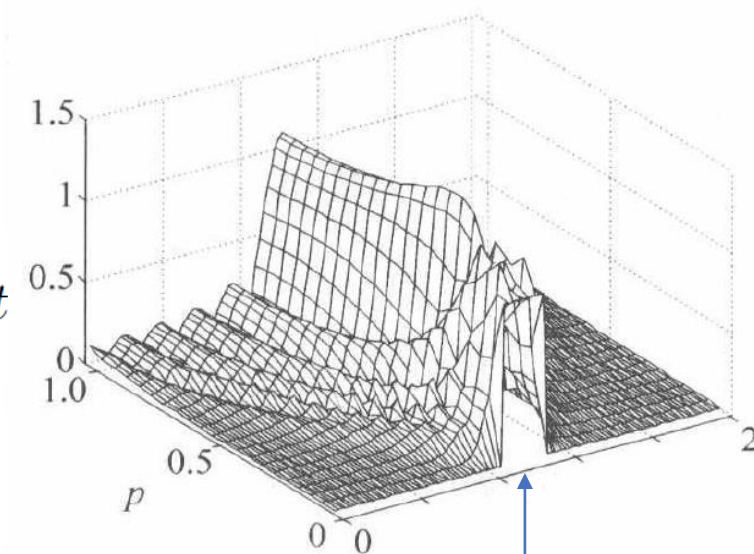
$$\mathcal{F}\{\psi_n(t)\} = \lambda_n \psi_n(t)$$

积分算符 $\mathcal{F}_p[f](x) = \int_{-\infty}^{\infty} K_p(x, t) f(t) dt$

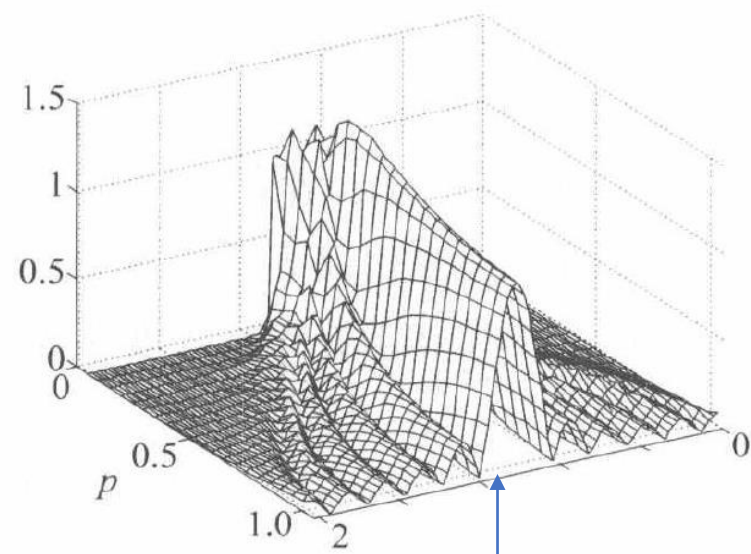
$$K_\alpha(x, t) = \sqrt{\frac{1 - i \cot \alpha}{2\pi}} \exp \left[i \frac{x^2 + t^2}{2} \cot \alpha - i \frac{xt}{\sin \alpha} \right]$$

$\alpha = p\pi/2 \downarrow p = 1$

$$K_{\pi/2}(x, t) = e^{-i\omega t}$$



方波



sinc

基本性质

线性

$$\mathcal{F}_\alpha[f](x) = \int_{-\infty}^{\infty} K_\alpha(x, t) f(t) dt$$

$$\mathcal{F}_\alpha[af(x) + bg(x)] = a \mathcal{F}_\alpha[f(x)] + b \mathcal{F}_\alpha[g(x)]$$

叠加性

$$\mathcal{F}_\alpha \circ \mathcal{F}_\beta = \mathcal{F}_{\alpha+\beta}$$

$$\mathcal{F}_\alpha(\mathcal{F}_\beta[\psi_n]) = e^{-in\beta} e^{-in\alpha} \psi_n = e^{-in(\alpha+\beta)} \psi_n = \mathcal{F}_{\alpha+\beta}[\psi_n]$$

Parseval 定理

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\mathcal{F}_\alpha[f](u)|^2 du$$

$$\int_{-\infty}^{\infty} f(x) g^*(x) dx = \int_{-\infty}^{\infty} \mathcal{F}_\alpha[f](u) \mathcal{F}_\alpha[g]^*(u) du.$$

Wigner 分布

$$W_f(x, \omega) = \int_{-\infty}^{\infty} f\left(x + \frac{\tau}{2}\right) f^*\left(x - \frac{\tau}{2}\right) e^{-i\omega\tau} d\tau$$

信号时间/空间-频率分布

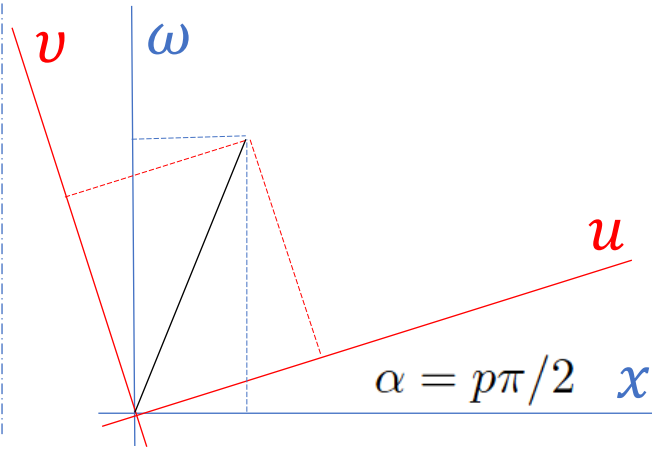
$$\int_{-\infty}^{\infty} W_f(x, \omega) d\omega = |f(x)|^2 \quad \text{时间/空间域功率谱}$$

$$\int_{-\infty}^{\infty} W_f(x, \omega) dx = |\hat{f}(\omega)|^2 \quad \text{频率域功率谱}$$

变换后

变换前

$$g(u) = \mathcal{F}_\alpha[f(x)] \quad W_g(u, \nu) = W_f(x, \omega) \Big|_{(x, \omega) = R_{-\alpha}(u, \nu)}$$



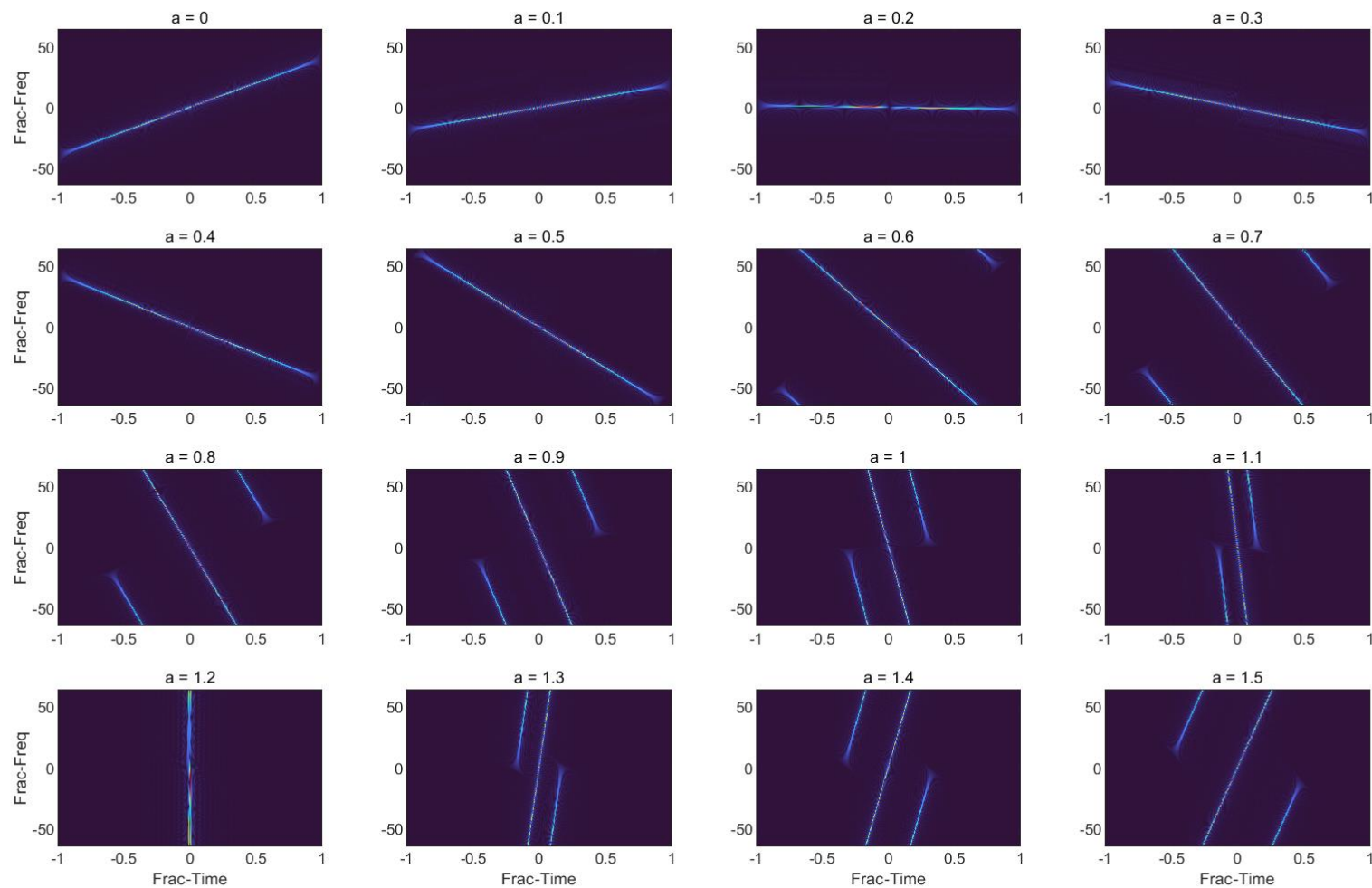
$$\begin{pmatrix} x \\ \omega \end{pmatrix} = R_{-\alpha} \begin{pmatrix} u \\ \nu \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} u \\ \nu \end{pmatrix}$$

几何意义

信号Wigner分布的旋转

基本性质

Wigner 分布随 FrFT 阶数 a 的旋转演变



基本性质

FT

处理基函数形式的信号

FrFT

$$f(t) = \exp(i\omega_0 t)$$

$$K(u, t) = \exp\{-iut\}$$

$$\mathcal{F}[f](u) = \int_{-\infty}^{\infty} e^{-iut} e^{i\omega_0 t} dt = 2\pi \delta(u - \omega_0)$$

恒定频率信号

—— FT选择出恒定频率

$$f(t) = \exp(i\pi a t^2) \quad \omega(t) = \frac{d}{dt}(\pi a t^2) = 2\pi a t$$
$$K_\alpha(u, t) = \sqrt{\frac{1 - i \cot \alpha}{2\pi}} \exp\left[i \frac{u^2 + t^2}{2} \cot \alpha - i \frac{ut}{\sin \alpha}\right]$$

$$\mathcal{F}_\alpha[f](u) = \int_{-\infty}^{\infty} K_\alpha(u, t) e^{i\pi a t^2} dt$$
$$= C_\alpha \int_{-\infty}^{\infty} \exp\left\{i \left[\left(\pi a + \frac{1}{2} \cot \alpha \right) t^2 - \frac{u}{\sin \alpha} t + \frac{u^2}{2} \cot \alpha \right] \right\} dt$$

$$\pi a + \frac{1}{2} \cot \alpha = 0 \quad \Longleftrightarrow \quad a = -\frac{1}{2\pi} \cot \alpha$$

$$\mathcal{F}_\alpha[f](u) \propto \int_{-\infty}^{\infty} e^{-i \frac{u}{\sin \alpha} t} dt = 2\pi \delta\left(\frac{u}{\sin \alpha}\right) \propto \delta(u)$$

线性调频信号

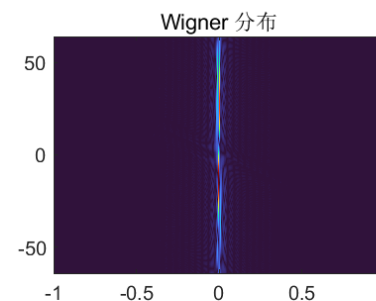
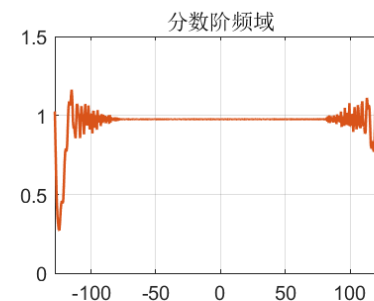
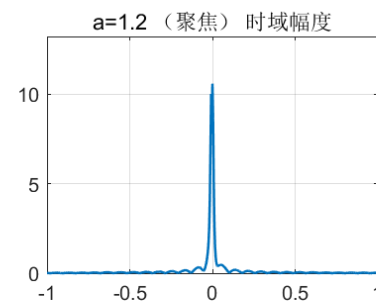
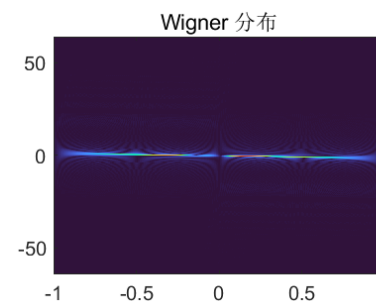
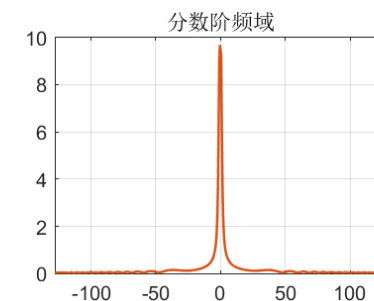
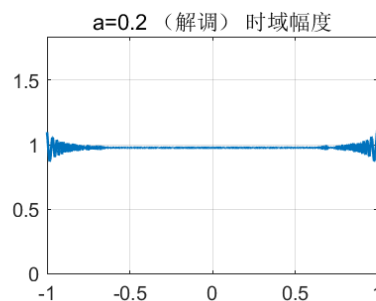
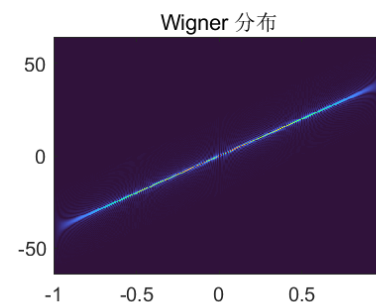
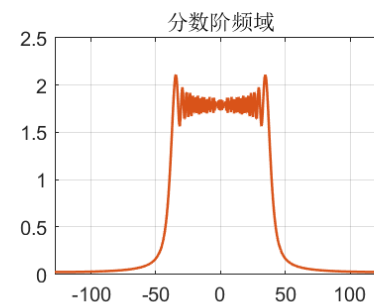
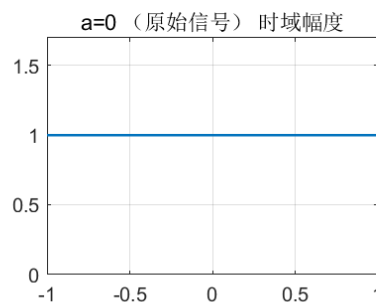
—— FrFT选择出Chrip信号

基本性质

幅度：
信号强度的模

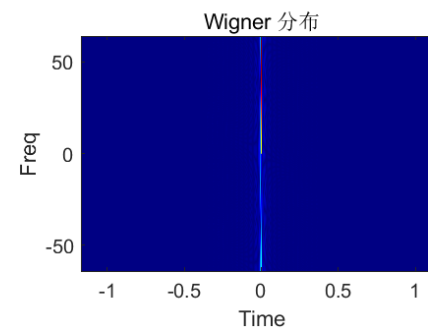
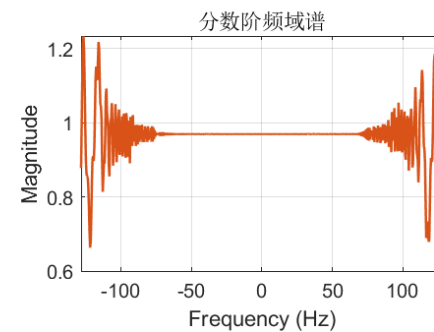
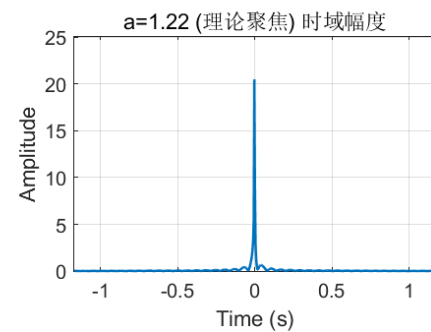
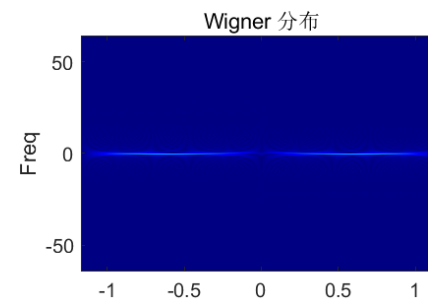
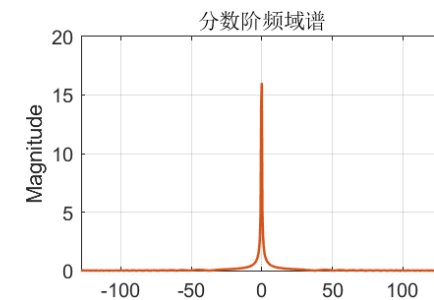
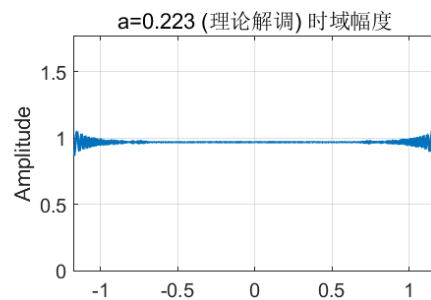
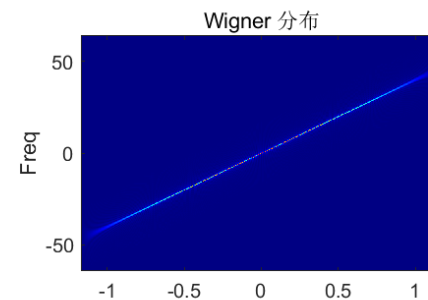
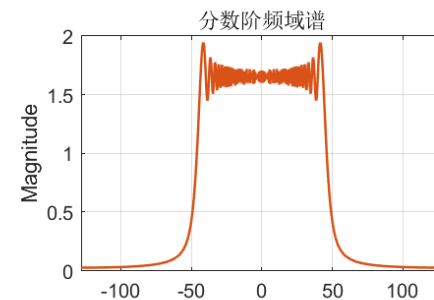
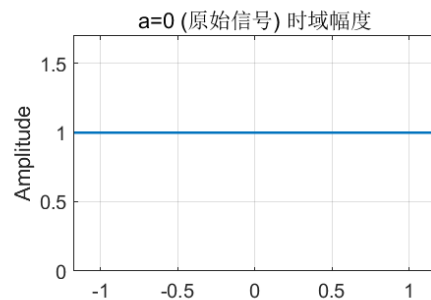
$$f(t) = \exp(i\pi at^2)$$

Chirp 信号

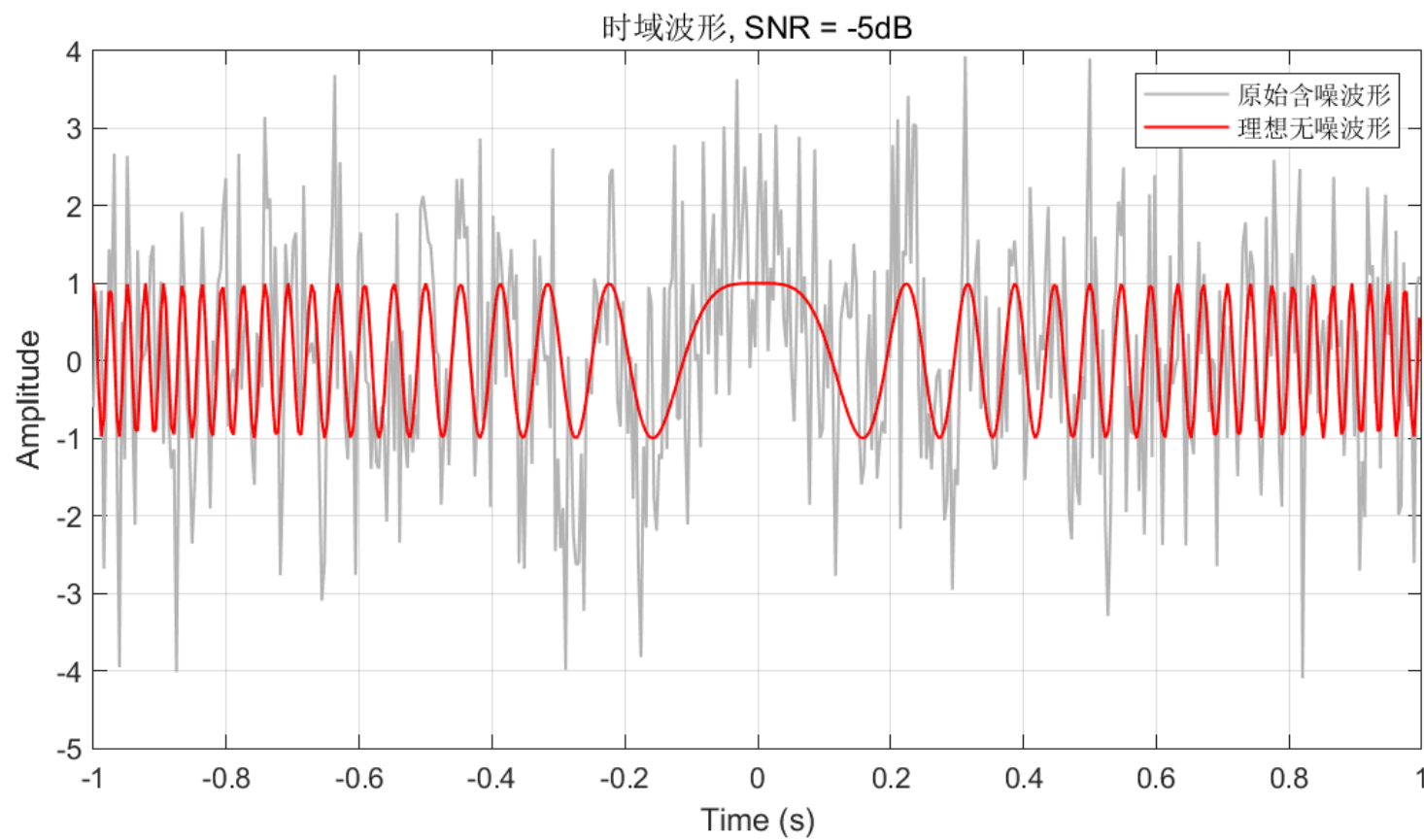


基本性质

FrFT阶数与Chirp匹配: $k = 40 \text{ Hz/s}$

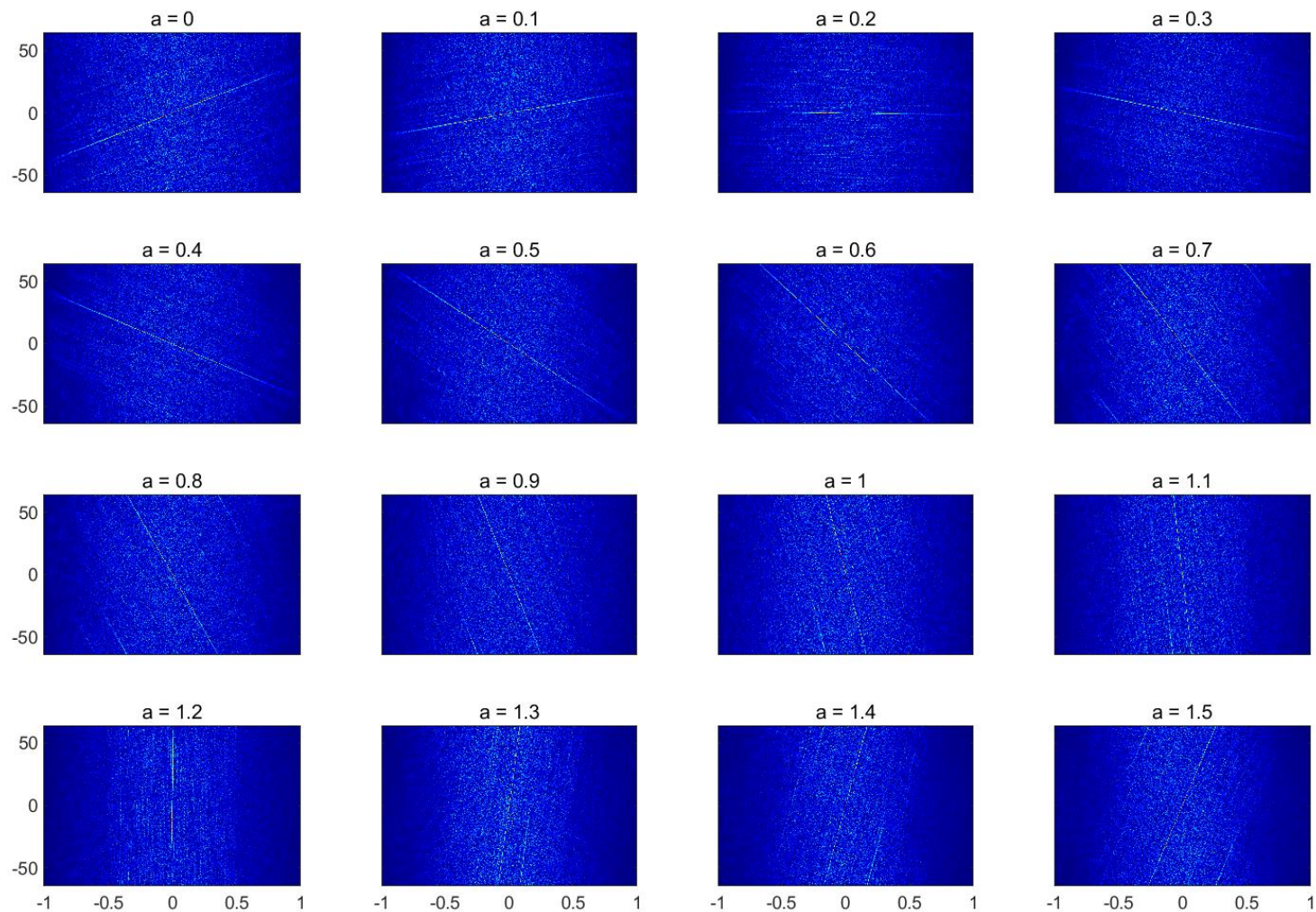


应用设想



应用设想

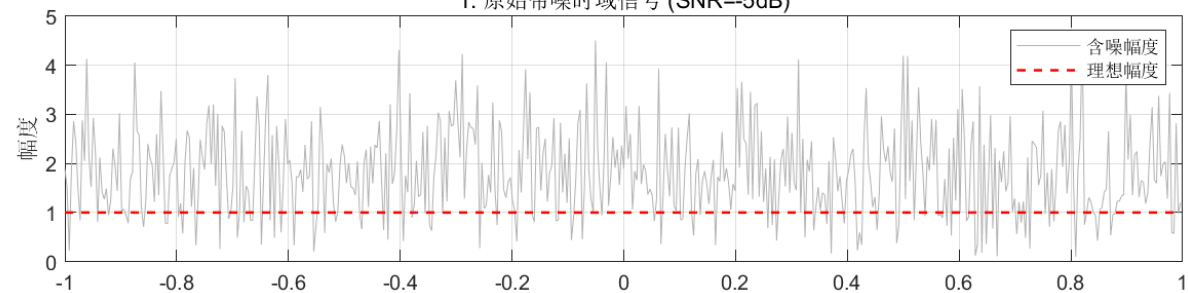
含噪信号的 Wigner 分布变化 (SNR = -5dB)



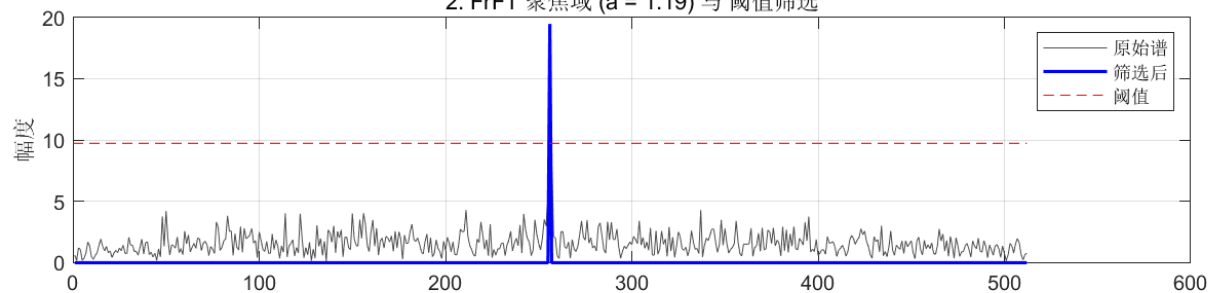
应用设想

基于 FrFT 聚焦滤波的信噪分离

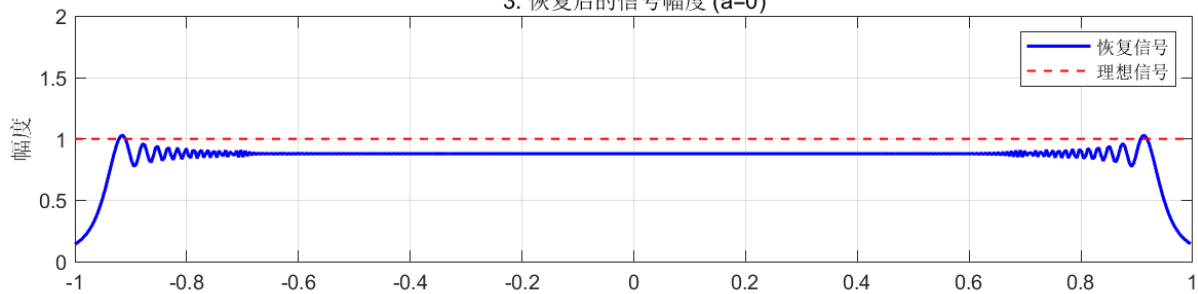
1. 原始带噪时域信号 (SNR=-5dB)



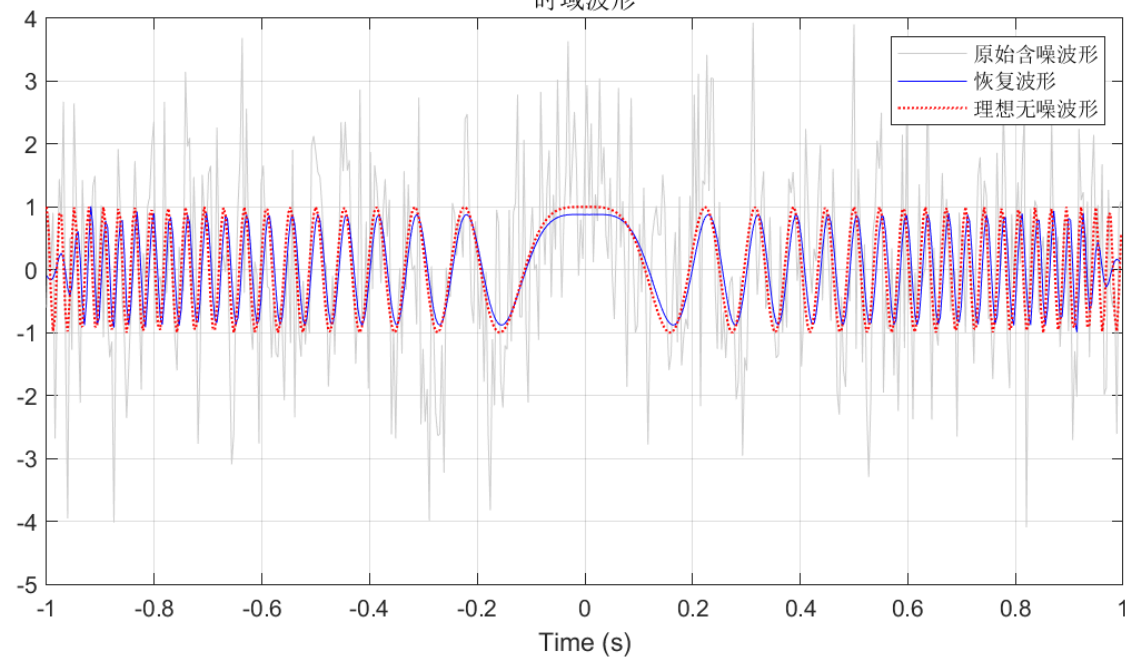
2. FrFT 聚焦域 ($a = 1.19$) 与 阈值筛选



3. 恢复后的信号幅度 ($a=0$)

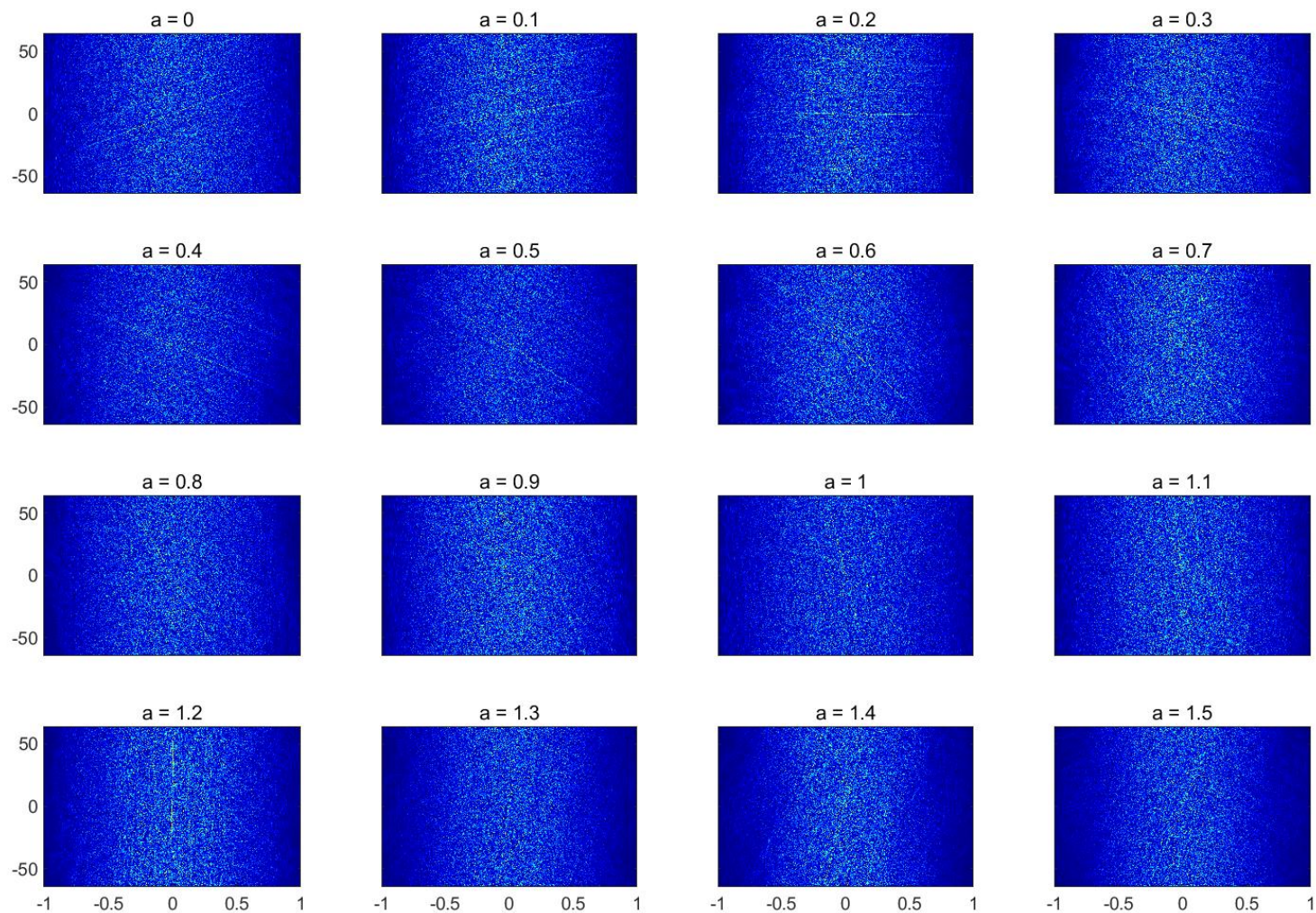


时域波形



应用设想

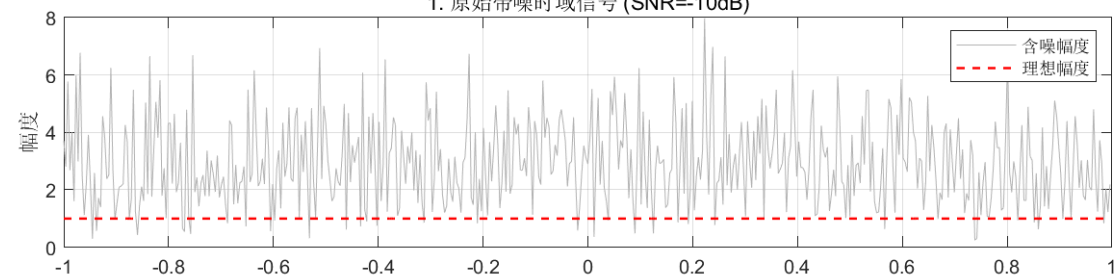
含噪信号的 Wigner 分布变化 (SNR = -10dB)



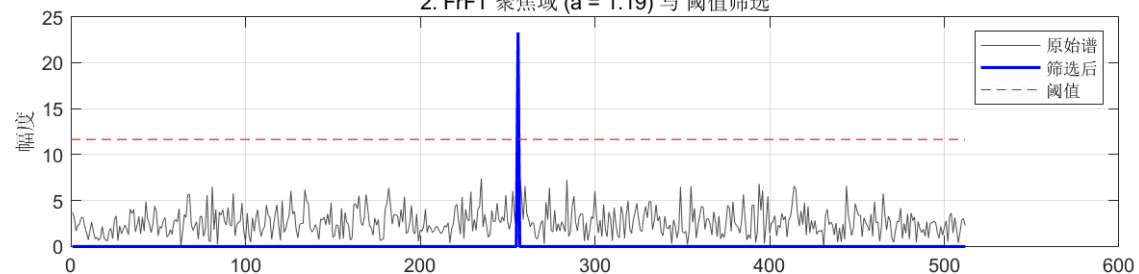
应用设想

基于 FrFT 聚焦滤波的信噪分离

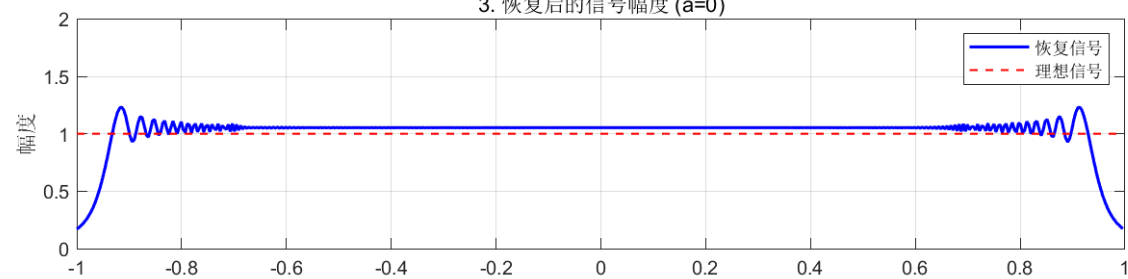
1. 原始带噪时域信号 (SNR=-10dB)



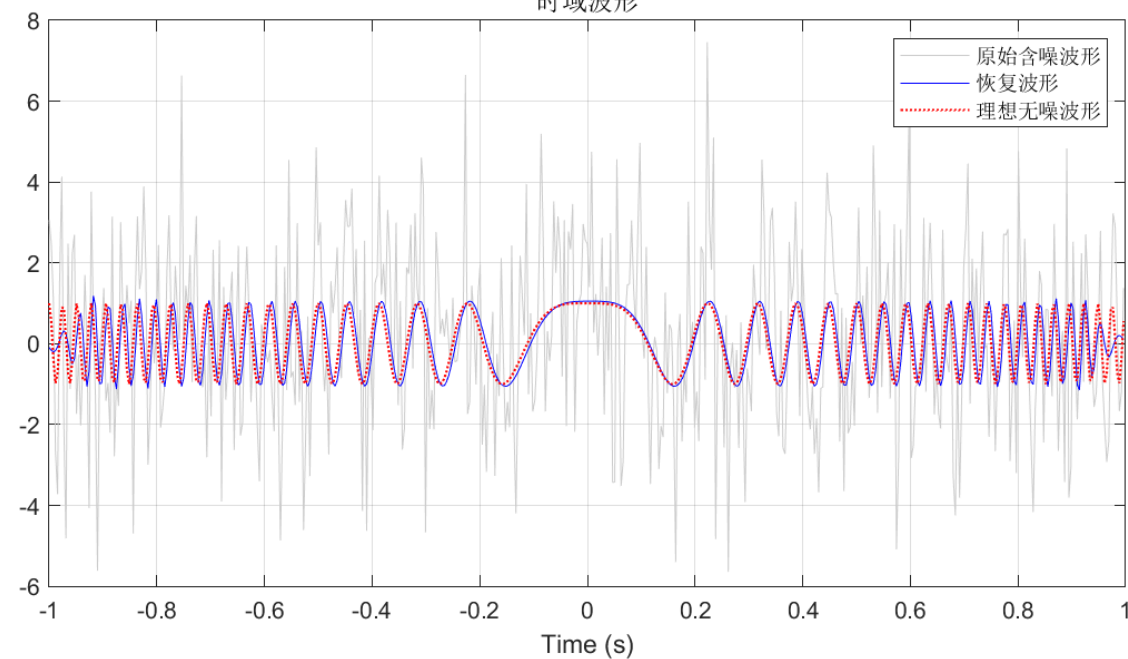
2. FrFT 聚焦域 ($a = 1.19$) 与 阈值筛选



3. 恢复后的信号幅度 ($a=0$)



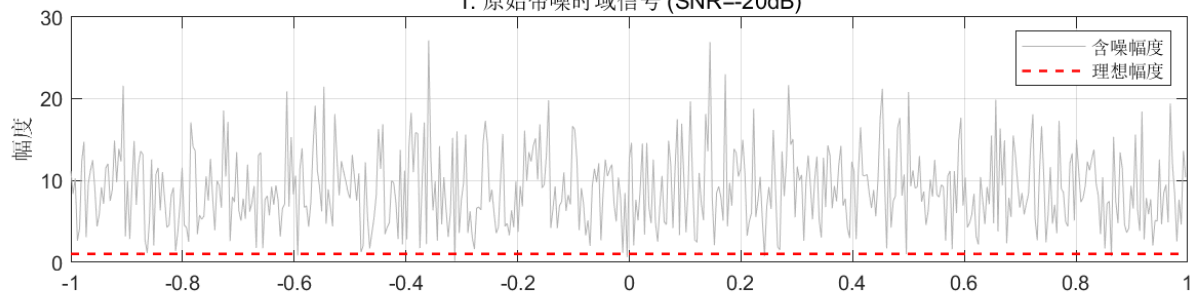
时域波形



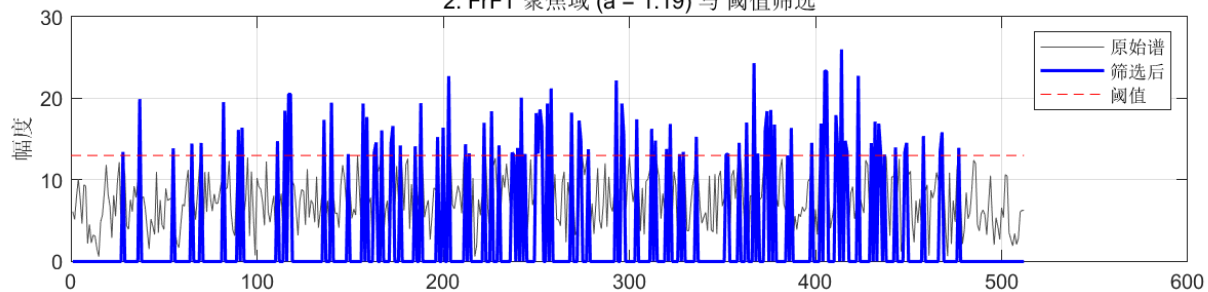
应用设想

基于 FrFT 聚焦滤波的信噪分离

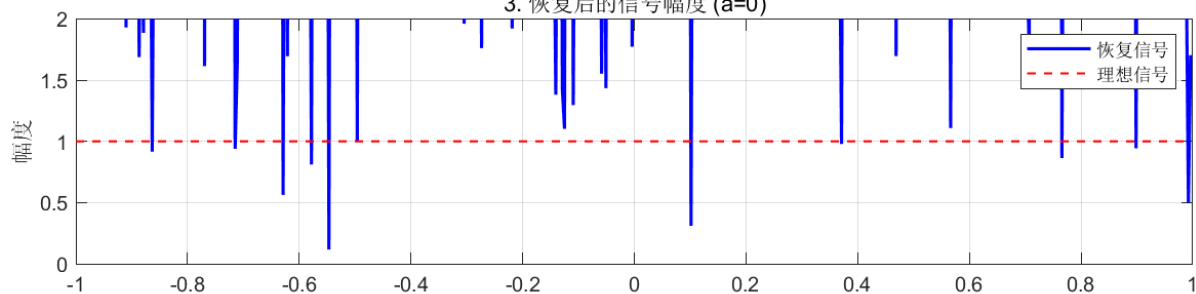
1. 原始带噪时域信号 (SNR=-20dB)



2. FrFT 聚焦域 ($a = 1.19$) 与 阈值筛选



3. 恢复后的信号幅度 ($a=0$)



时域波形

